Daniel Jang

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Technical Skills

Languages: Java, C, Python, JavaScript, TypeScript, Dart, HTML/CSS, SQL, Ruby
Developer Tools: Git, Docker, GCP, VS Code, Bash, Linux, Maven, Powershell
Frameworks: React, Node.js, Spring Boot, Flutter, Rails, Flask, Vite, Gemini API

Projects

Siel – Digital Biographer | *Java 17, Spring Boot, Node.js, Powershell, Gemini API* — May 2026

- Designed a photo-driven journaling platform, using in-memory processing to extract metadata without committing sensitive media to disk, eliminating the local data footprint.
- Developed a high-concurrency metadata pipeline in **Java 17** using **ParallelStreams** to execute multi-threaded EXIF extraction, significantly reducing processing time for bulk image batches.
- Hardened system security via **Argon2id** hashing and **RS256** asymmetric JWT signing, integrating the **Gemini API** to generate journal narratives within a secure environment.

Mappin' – Destination Alarm System | *TypeScript, JavaScript, HTML, CSS* — Apr. 2026

- Constructed a cross-platform location alarm that notifies users upon arrival at their intended destinations by implementing **Service Worker logic** between a **Chrome Extension (MV3)** and a **PWA**.
- Optimized data extraction by implementing a URL parser that uses **URL APIs** to resolve shortened redirects, successfully mapping destination coordinates across **30+ varying formats** with a **0% failure rate**.
- Implemented a high-precision arrival detection system by integrating the **Haversine formula**, allowing the model to calculate real-time optimal distance and trigger synchronized **haptic alerts** across user-defined radii from **50m to 2000m**.

UCEY – Urban Planning Interface | *TypeScript, React, Python, Web APIs* — Mar. 2026

- Developed a web app that helps urban planners find underused city land like brownfields and estimate the redevelopment cost using **TypeScript** and **React**.
- Built the full stack design, connecting **map APIs** and the **figures from Statistics Canada** to the interactive interface so users can accurately browse and assess sites directly on a map instead of manually looking through raw data.
- Strengthened platform security by including an authentication system featuring **MFA** and **secure tokens** using APIs to ensure protected, personalized user sessions validated by a successful **pilot test with 5+ users**.

YSafe – Pedestrian Safety Navigation App | *Dart, Google Maps API, ArcGIS, Hive* — Feb. 2026

- Designed a navigation web app that recommends routes based on localized safety scores atop simple distance optimization by integrating **Flutter** with the **Google APIs**, implementing a custom weighting logic.
- Optimized route-generation by extracting **5 years of York Region Open Data**, allowing the app to recommend routes based on historically occurred safety incidents.
- Improved performance using **Riverpod** framework and **Hive** caching, enabling an alert engine to deliver real-time navigation warnings.

Experience & Activities

Waterloo Rocketry — Nov. 2025 – Present
Software Member — *Waterloo, ON*

- Analyzed data collection workflows using **Python** and **Java**, ensuring software stability during rocket launch operations.
- Collaborated with the team to develop a full-stack, ground control software using **TypeScript** and **Python**.
- Handling feature stabilization for the July architecture lock, focusing on integration testing and debugging to ensure software stability for the upcoming launch.

Waterloo Aerial Robotics Group — Nov. 2025 – Present
Autonomy Member — *Waterloo, ON*

- Testing autonomous flight performance by engineering a data pipeline for object detection and using the flight-test datasets to analyze the model training accuracy.
- Bridged the gap between the camera and ground control by developing a real-time data stream that processes detected targets, enabling the team to verify autonomous tasks.

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science, Honours | Term Distinction - 1B — *Sep. 2025 – Apr. 2030*

- **Relevant Coursework:** Algorithms, Memory Management, Shell Scripting, Data Structures, OOP, Linear Algebra, Calculus II, Probability